

**Ime in Priimek:** \_\_\_\_\_**Naloga 1**

Zapiši s kotnimi funkcijami ostrih kotov v kotnih stopinjah in izračunaj:

a) 
$$\frac{\cos 405^\circ + \sin(-150^\circ)}{\tan(-765^\circ) \cdot \cot(-240^\circ)}$$

b) 
$$\frac{\sin(1110^\circ - \cos 495^\circ)}{\tan(-675^\circ) + \cot^2 300^\circ}$$

c) 
$$\frac{\cos(-300^\circ) \cdot \sin 480^\circ}{\tan 1140^\circ - \cot 135^\circ}$$

d) 
$$\frac{\cos 1080^\circ - \sin(-570^\circ)}{\frac{1}{2} \tan 225^\circ + \cot(-390^\circ)} + \sin^2(-30^\circ)$$

e) 
$$\frac{\sin^2 750^\circ + \cos 225^\circ}{\tan(-570^\circ) \cdot \cot 330^\circ}$$

## Naloga 2

Zapiši s kotnimi funkcijami ostrih kotov v radianih in izračunaj:

$$\text{a) } \frac{\sin \frac{17\pi}{4} + \cos \left(-\frac{5\pi}{3}\right)}{\tan \left(-\frac{9\pi}{2}\right) \cdot \cot^2 \left(\frac{5\pi}{6}\right)}$$

$$\text{d) } \frac{\cos \left(-\frac{10\pi}{3}\right) + \sin \frac{23\pi}{4}}{\tan \frac{17\pi}{6} \cdot \cot \left(-\frac{5\pi}{4}\right)} + \sin \frac{13\pi}{6}$$

$$\text{b) } \frac{\cos \frac{25\pi}{6} - \sin \left(-\frac{7\pi}{4}\right)}{\tan \frac{19\pi}{3} + \cot \frac{3\pi}{4}} \cdot \left(\tan \frac{4\pi}{3} - \cot \frac{5\pi}{4}\right)$$

$$\text{e) } \frac{\sin \frac{29\pi}{6} - \cos \frac{19\pi}{4}}{\tan \left(-\frac{11\pi}{3}\right) + \cot^2 \frac{7\pi}{4}}$$

$$\text{c) } \frac{\sin^2 \frac{13\pi}{3} \cdot \cos \left(-\frac{11\pi}{6}\right)}{\tan \frac{15\pi}{4} - \cot \frac{4\pi}{3}}$$

## Naloga 3

Zavrti točko  $A\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$  za pravi kot okoli koordinatnega izhodišča v smeri urnega kazalca. Določi koordinate te točke.

## Naloga 4

Na krožnici s polmerom 2 leži točka  $T(\sqrt{3}, y < 0)$ .

a) Kakšni koordinati ima ta točka, če a) jo zavrtimo okoli koordinatnega izhodišča v nasprotni smeri gibanja urnega kazalca za kot  $165^\circ$ ?

b) če bi jo zavrteli v smeri gibanja urnega kazalca za kot  $450^\circ$ ?

## Naloga 5

Poenostavi:

a)  $(1 + \tan^2 x)(1 - \sin^2 x)$

b)  $(1 + \cot^2 x)(1 - \cos^2 x)$

c)  $(3 \cos x + 2 \sin x)^2 + (3 \cos x - 2 \sin x)^2$

d)  $\frac{1 + \cos x}{\sin x} - \frac{\sin x}{1 - \cos x}$

e)  $\sin\left(x + \frac{\pi}{3}\right) + \sin\left(x - \frac{7\pi}{3}\right)$

f)  $\cos\left(x + \frac{\pi}{4}\right) + \cos\left(x - \frac{9\pi}{4}\right)$

g)  $\tan\left(x + \frac{\pi}{4}\right) \cdot \tan\left(x - \frac{5\pi}{4}\right)$

## Naloga 6

Poenostavi:

a)  $\frac{\sin 2x}{1 + \cos 2x}$

b)  $\frac{\cot^2 x - 1}{2 \cot x} \cdot \frac{\sin 2x}{2 \cos^2 x - 1}$

c)  $\cot x - \tan x - 2 \cot 2x$

## Naloga 7

a) Naj velja za oster kot  $x$ :  $\tan x = \frac{12}{5}$ . Izračunaj  $\sin 2x$  in  $\cos(x - \frac{\pi}{4})$ .

b) Naj bo  $0 < x < \pi$  in  $\cot x = \frac{3}{4}$ . Izračunaj  $\cos 2x$  in  $\tan(x - \frac{\pi}{4})$ .

c) Naj bo  $\pi < x < 2\pi$  in  $\cos x = \frac{\sqrt{5}}{3}$ . Izračunaj  $\tan 2x$  in  $\cot(x + \frac{\pi}{4})$ .

d) Naj bo  $x$  topi kot in  $\sin x = \frac{\sqrt{15}}{4}$ . Izračunaj  $\sin(x + \frac{\pi}{4})$  in  $\cos(x - \frac{5\pi}{4})$ .

e) Naj za ostru kota  $x$  in  $y$  velja  $\sin x = \frac{3}{5}$  in  $\cos y = \frac{5}{13}$ . Izračunaj  $\tan(x - y)$ ,  $\sin \frac{x}{2}$ ,  $\tan \frac{y}{2}$ .

f) Naj bo  $\frac{3\pi}{2} < x < 2\pi$  in  $\sin 2x = -\frac{24}{25}$ . Izračunaj  $\sin x$ ,  $\sin \frac{x}{2}$ .

g) Naj bo  $\frac{\pi}{2} < x < \pi$  in  $\cos 2x = \frac{40}{41}$ . Izračunaj  $\cos x$ ,  $\cos \frac{x}{2}$ .

h) Naj bo  $x$  oster kot in  $\tan 2x = \frac{7}{24}$ . Izračunaj  $\tan x$ ,  $\tan \frac{x}{2}$ .

## Naloga 8

Dokaži identiteto:

a)  $\frac{\sin(3x)}{\sin x} - \frac{\cos(3x)}{\cos x} = 2$

b)  $\sin^4 x = \frac{3 - 4 \cos(2x) + \cos(4x)}{8}$

c)  $\tan(3x) = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$

d)  $\cos 4x = 8 \cos^4 x - 8 \cos^2 x + 1$

### Naloga 9

Dokaži enakost:

a)  $\frac{\cos x \sin 2x + \cos 2x + 1}{\sin x + 1} \cdot (1 + \tan^2 x) = 2$

b)  $\frac{\cos x + \cos 2x + 1}{\sin x + \sin 2x} = \cot x$

c)  $\frac{\sin 2x}{\sin x} (1 + \tan^2 x) = \frac{2}{\cos x}$

### Naloga 10

Zapiši kot produkt in izračunaj:

a)  $\cos 105^\circ - \cos 15^\circ$

d)  $\sin 5^\circ + \sin 55^\circ - \cos 25^\circ$

b)  $\sin 105^\circ + \sin 15^\circ$

e)  $\sin 10^\circ + \sin 50^\circ - \cos 20^\circ$

c)  $\cos 15^\circ - \sin 15^\circ$

f)  $\sin 8^\circ + \cos 38^\circ + \cos 158^\circ$

### Naloga 11

Zapiši kot vsoto ali razliko:

a)  $\cos 7x \cos 5x$

d)  $\sin(2x + 3y) \sin(2x - 3y)$

b)  $\sin 2x \sin 7x$

e)  $\cos(x + y) \cos(x - y)$

c)  $\sin 6x \cos 4x$

f)  $2 \sin(4x + y) \cos(4x - y)$

### Naloga 12

Pokaži, da v trikotniku velja:  $\sin 2\alpha + \sin 2\beta + \sin 2\gamma = 4 \sin \alpha \sin \beta \sin \gamma$

### Naloga 13

Dokaži enakost:

a)  $\frac{2 \sin 5x \cdot \sin 4x}{\cos x - \cos 9x} = 1$

b)  $\frac{\cos 2x \cos 10x}{\cos 12x + \cos 8x} = \frac{1}{2}$

c)  $\frac{\sin 4x + \sin 8x}{\cos 4x - \cos 8x} = \cot 2x$

## Naloga 14

Nariši grafe funkcij:

a)  $f(x) = 2 \cos(x + \frac{\pi}{4}) - 1$

b)  $f(x) = -2 \sin(x + \pi) - 1$

c)  $f(x) = \tan\left(\frac{x}{2} - \frac{\pi}{4}\right)$

a)  $f(x) = |\cos(x - \frac{\pi}{3})| - 1$

b)  $f(x) = -|\sin\left(\frac{x}{2} + \frac{\pi}{4}\right) + 2|$

c)  $f(x) = |\cot\left(\frac{x}{3} - \frac{\pi}{6}\right)|$

## Naloga 15

Reši enačbo:

a)  $\cos(x + \frac{\pi}{2}) = \frac{\sqrt{2}}{2}$

b)  $2 \sin(x + \pi) = -\sqrt{3}$

c)  $-3 \tan(\frac{x}{3} + \frac{\pi}{6}) = \sqrt{3}$

d)  $\cot(\frac{x}{2} + \frac{\pi}{3}) = \sqrt{3}$

e)  $\cot^2 x - \frac{1}{9} = 0$

f)  $\sin^2 x - \cos x = 1$

g)  $\sin^2 x - \sin x = 2$

h)  $\sin x = \sqrt{3} \cos x$

i)  $\sin 2x \tan x = \frac{3}{2}$

j)  $\cos 3x \sin 2x - \cos 3x = 0$

k)  $\sin^2 x + 2 \sin x \cos x - 3 \cos^2 x = 1$

l)  $\sin^2 x - 6 \sin x \cos x - \cos^2 x = 3$

m)  $\cos 2x = 5 \sin^2 x - \cos^2 x$

n)  $\cos^3 x - \cos x = \sin^2 x$

1. Kote prevedemo v osnovne:  $-405^\circ = 360^\circ + 45^\circ \Rightarrow \cos(405^\circ) = \cos(45^\circ) = \frac{\sqrt{2}}{2}$  -  $\sin(-150^\circ) = -\sin(150^\circ) = -\sin(180^\circ - 30^\circ) = -\sin(30^\circ) = -\frac{1}{2}$  -  $765^\circ = 2 \cdot 360^\circ + 45^\circ \Rightarrow \tan(765^\circ) = \tan(45^\circ) = 1$  -  $240^\circ = 180^\circ + 60^\circ \Rightarrow \cot(240^\circ) = \cot(60^\circ) = \frac{\sqrt{3}}{3}$

Števec:  $\frac{\sqrt{2}}{2} + \left(-\frac{1}{2}\right) = \frac{\sqrt{2}-1}{2}$

Imenovalec:  $1 \cdot \frac{\sqrt{3}}{3} = \frac{\sqrt{3}}{3}$

Celoten izraz:

$$\frac{\frac{\sqrt{2}-1}{2}}{\frac{\sqrt{3}}{3}} = \frac{\sqrt{2}-1}{2} \cdot \frac{3}{\sqrt{3}} = \frac{3(\sqrt{2}-1)}{2\sqrt{3}} = \frac{\sqrt{3}(\sqrt{2}-1)}{2}$$

$$\boxed{\frac{\sqrt{3}(\sqrt{2}-1)}{2}}$$

2. Kote prevedemo:  $-1110^\circ = 3 \cdot 360^\circ + 30^\circ \Rightarrow \sin(1110^\circ) = \sin(30^\circ) = \frac{1}{2}$  -  $495^\circ = 360^\circ + 135^\circ \Rightarrow \cos(495^\circ) = \cos(135^\circ) = -\frac{\sqrt{2}}{2}$  -  $-675^\circ = -2 \cdot 360^\circ + 45^\circ \Rightarrow \tan(-675^\circ) = \tan(45^\circ) = 1$  -  $300^\circ = 360^\circ - 60^\circ \Rightarrow \cot(300^\circ) = -\cot(60^\circ) = -\frac{\sqrt{3}}{3} \Rightarrow \cot^2(300^\circ) = \frac{1}{3}$

Števec:  $\frac{1}{2} - \left(-\frac{\sqrt{2}}{2}\right) = \frac{1+\sqrt{2}}{2}$

Imenovalec:  $1 + \frac{1}{3} = \frac{4}{3}$

Celoten izraz:

$$\frac{\frac{1+\sqrt{2}}{2}}{\frac{4}{3}} = \frac{1+\sqrt{2}}{2} \cdot \frac{3}{4} = \frac{3(1+\sqrt{2})}{8}$$

$$\boxed{\frac{3(1+\sqrt{2})}{8}}$$

3. Kote prevedemo:  $-\cos(-300^\circ) = \cos(300^\circ) = \cos(360^\circ - 60^\circ) = \cos(60^\circ) = \frac{1}{2}$  -  $480^\circ = 360^\circ + 120^\circ \Rightarrow \sin(480^\circ) = \sin(120^\circ) = \sin(60^\circ) = \frac{\sqrt{3}}{2}$  -  $1140^\circ = 3 \cdot 360^\circ + 60^\circ \Rightarrow \tan(1140^\circ) = \tan(60^\circ) = \sqrt{3}$  -  $135^\circ = 180^\circ - 45^\circ \Rightarrow \cot(135^\circ) = -\cot(45^\circ) = -1$

Števec:  $\frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$

Imenovalec:  $\sqrt{3} - (-1) = \sqrt{3} + 1$

Celoten izraz:

$$\frac{\frac{\sqrt{3}}{4}}{\sqrt{3}+1} = \frac{\sqrt{3}}{4(\sqrt{3}+1)} \cdot \frac{\sqrt{3}-1}{\sqrt{3}-1} = \frac{3-\sqrt{3}}{4(3-1)} = \frac{3-\sqrt{3}}{8}$$

$$\boxed{\frac{3-\sqrt{3}}{8}}$$

4. Kote prevedemo:  $-750^\circ = 2 \cdot 360^\circ + 30^\circ \Rightarrow \sin(750^\circ) = \sin(30^\circ) = \frac{1}{2} \Rightarrow \sin^2(750^\circ) = \frac{1}{4}$  -  $225^\circ = 180^\circ + 45^\circ \Rightarrow \cos(225^\circ) = -\cos(45^\circ) = -\frac{\sqrt{2}}{2}$  -  $-540^\circ = -2 \cdot 360^\circ + 180^\circ \Rightarrow \tan(-540^\circ) = \tan(180^\circ) = 0$  -  $330^\circ = 360^\circ - 30^\circ \Rightarrow \cot(330^\circ) = -\cot(30^\circ) = -\sqrt{3}$

Števec:  $\frac{1}{4} + \left(-\frac{\sqrt{2}}{2}\right) = \frac{1-2\sqrt{2}}{4}$

Imenovalec:  $0 \cdot (-\sqrt{3}) = 0$

Deljenje z nič je nedefinirano.

Nedefinirano

5. Kote prevedemo:  $-1080^\circ = 3 \cdot 360^\circ + 0^\circ \Rightarrow \cos(1080^\circ) = \cos(0^\circ) = 1$  -  $\sin(-570^\circ) = -\sin(570^\circ) = -\sin(360^\circ + 210^\circ) = -\sin(210^\circ) = -(-\sin(30^\circ)) = \frac{1}{2}$  -  $150^\circ = 180^\circ - 30^\circ \Rightarrow \tan(150^\circ) = -\tan(30^\circ) = -\frac{\sqrt{3}}{3}$  -  $420^\circ = 360^\circ + 60^\circ \Rightarrow \cot(420^\circ) = \cot(60^\circ) = \frac{\sqrt{3}}{3} \Rightarrow \cot^2(420^\circ) = \frac{1}{3}$

Števec:  $1 - \frac{1}{2} = \frac{1}{2}$

Imenovalec:  $-\frac{\sqrt{3}}{3} + \frac{1}{3} = \frac{1 - \sqrt{3}}{3}$

Celoten izraz:

$$\frac{\frac{1}{2}}{\frac{1-\sqrt{3}}{3}} = \frac{1}{2} \cdot \frac{3}{1-\sqrt{3}} = \frac{3}{2(1-\sqrt{3})} \cdot \frac{1+\sqrt{3}}{1+\sqrt{3}} = \frac{3(1+\sqrt{3})}{2(1-3)} = \frac{3(1+\sqrt{3})}{-4} = -\frac{3(1+\sqrt{3})}{4}$$

$$-\frac{3(1+\sqrt{3})}{4}$$